

Temporal Analysis Software

Technical and user documentation

Version Beta 0.1.2

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1. Introduction

The Temporal Analysis Software is a desktop tool for analyzing the hourly distribution of events (crimes, interventions, incidents, etc.) from a spreadsheet of data. Its main output is the **Data Clock**: a 7-day × 24-hour matrix that uses a color gradient to show which days and hours concentrate the analyzed events.

Besides the Data Clock, the software generates charts by time band and by day of the week, up to four counting analyses (modality, offense, unit and zone) and a general summary page, exportable to Excel, PDF and PNG.

2. Workflow

The program is organized into 5 steps, in the left sidebar:

- **1. Import:** choose the data file (.xlsx, .xls, .csv or .dbf).
- **2. Columns:** state which column of the file corresponds to each field the program needs (start/end date and time, unit, modality, zone, DelitoCOP).
- **3. Filters:** optionally narrow the analysis by date/time and by up to 3 more columns, each with its own values to include.
- **4. Settings:** title, color, time bands, and which optional analyses to generate.
- **5. Results:** enabled once the analysis is generated; shows every page of the report and lets you export it.

The **Run Analysis** button (bottom-right of "4. Settings", or in the top menu) processes the data with the chosen columns, filters and settings, and builds every page of the report.

3. Supported file formats

- **Excel** (.xlsx, .xls): the first sheet of the file is read.
- **CSV** (.csv): separator auto-detected (comma or semicolon).
- **DBF** (.dbf): dBase format, used by some older police/GIS systems. Read with a built-in decoder, no external library required. "Memo" fields (which require a separate .dbt file) are left blank.

4. Required columns and date/time format

In step "2. Columns", all 8 of these columns must be set: start date, start time, end date, end time, unit, modality, zone and DelitoCOP.

Automatic date/time detection: by default, the program tries several common formats (YYYY-MM-DD, DD/MM/YYYY, HH:MM:SS, etc.) until one works with the file's data.

Manual format: if automatic detection fails (for example, with ambiguous dates like "03/04/2026", which could be April 3rd or March 4th), check "Manually choose the date and time format" and pick the exact format from a list of options (including AM/PM formats). That format is tried before the automatic ones.

Midnight crossing without a real end-date column: if the end time is earlier than the start time AND the start-date and end-date columns are the same column in the file, the program assumes the event ended the next day and adds one day to the end date automatically. If the end time equals the start time, it's treated as an instant event (not as having lasted 24 hours).

5. Filters

Date/time filter: narrows the analysis to a specific date/time range.

Column filters (up to 3, independent of each other): each one picks a column from the file and lists its unique values, all checked by default. Unchecking some values excludes from the analysis any events that have them.

Text search: each filter has a search box. Typing text there (e.g. "moto") automatically checks only the values that contain it ("MOTOS(ROBO)", "MOTOCHORROS", etc.) and unchecks the rest — it's not just a visual magnifying glass, it changes the actual selection. Clearing the search shows every value again, with whatever check state was left.

Combining several filters at once (e.g. date + unit + modality) works as a logical AND: only events that meet every condition at the same time remain.

6. How hours are computed — the heart of the Data Clock

This is the most important part for correctly reading the Data Clock's numbers. **Each event contributes a total of exactly 1** to the matrix, no matter how long it lasted. That "1" is split proportionally across every hour the event occupies.

- **Instant event** (start time equals end time, e.g. an incident logged "at 14:00"): adds a full 1 to that single hour.
- **Event with a duration, within a single time span** (e.g. from 11:20 to 12:40): the 1 is split proportionally to the time spent in each clock hour it touches. In this example, it spends 40 minutes in hour 11 and 40 minutes in hour 12 (out of an 80-minute total duration), so it adds 0.50 to hour 11 and 0.50 to hour 12.
- **Event spanning several full hours** (e.g. from 10:00 to 13:00, 3 hours): the 1 is split evenly across those 3 hours (0.33 each).

Event	Duration	Hour 11	Hour 12	Total
11:20 to 12:40	1h 20min	0.50	0.50	1.00
11:00 to 11:00 (instant)	0 min	1.00	0.00	1.00
10:00 to 13:00	3h	0.33 (at h.10)*	0.33	1.00 (split across 3 hours)

*Values are rounded to 2 decimals for the table; internally they're computed with more precision.

Thanks to this proportional split, the **sum of every cell in the matrix always equals the number of valid events analyzed** (no more, no less), regardless of how long each one lasted. This is what makes the grand total (the "FREQ" cell at the bottom right) useful as a cross-check.

Midnight crossing: if an event crosses midnight (e.g. from 23:00 to 01:00), each hour is counted on the day it actually belongs to: hour 23 on the day it started, and hours 0 and 1 on the next day.

Events longer than 24 hours: treated as a data-entry error (e.g. a mistyped end date) and are NOT included in the matrix. They're counted separately, as "excluded", and that count is computed over the events that already passed the chosen filters (not over the whole file): if you filter by a given unit and 5 of those filtered events exceed 24 hours, the program shows exactly those 5 as excluded, not however many exist in the entire file.

Date/time format errors: rows that can't be parsed as a valid date/time (e.g. an empty cell or unrecognizable text) are also left out of the analysis, and counted separately as "format errors".

7. Configurable time bands

Besides the hour-by-hour matrix, the Data Clock groups the 24 hours into wider bands (2, 3 or 4, your choice) to read the overall pattern at a glance.

- **Number of bands:** 4 bands of 6 hours, 3 of 8, or 2 of 12.
- **Start hour of the first band:** configurable (00:00 by default). For example, with a 07:00 start hour and 4 bands: 07-13, 13-19, 19-01 and 01-07.
- The **00-to-23 matrix never changes layout** because of this: only how those 24 columns are grouped for the totals-by-band block, drawn below, changes. If a band crosses midnight, it's drawn split into 2 segments (e.g. 19-23 and 00-01), but it still counts as a single band for calculation purposes.

8. Counting analyses (Modality, Offense, Unit, Zone)

Four optional analyses, independent of each other (any combination can be enabled, even all 4 at once). Each builds a *ranking*: it counts how many times each distinct value of the chosen column appears (among the events that passed the filters), sorted from highest to lowest frequency.

Separator (Modality only): sometimes a single "modality" cell holds more than one value together, separated by a comma (e.g. "MOTOCHORRO, FIREARM"). With the separator set ("," by default), the program counts each value separately: that example adds 1 to "MOTOCHORRO" and 1 to "FIREARM", not 1 to the whole combination. The other analyses (Offense, Unit, Zone) don't use a separator: each cell counts as a single, whole value.

Percentage: each value's frequency divided by the total number of occurrences counted in that analysis (not by the number of events, if a separator was used and some cells held more than one value).

Amount to chart: how many ranking values to show in that page's chart and table (0 = all). The exported Excel always includes the full ranking, even if the PDF/screen version was limited to fewer items for space.

Each enabled analysis adds its own page to the report (table on top, bar chart below), so it never competes for space with the others.

9. General Summary

Optional page (checkbox in "4. Settings") meant for a quick look without going through the whole report. Shows, as cards: events analyzed, peak day, peak hour, peak time band, and the top value (the first one in the ranking) of each enabled counting analysis. Below that, the charts by time band and by day of the week, and a "Top 3" for each enabled analysis.

- **Peak day/hour:** the day of the week and the hour of the day with the highest cumulative frequency in the Data Clock matrix.
- **Peak time band:** the time band (from the ones set up in step 4) with the highest frequency.

10. Visual customization

- **Title and subtitle** of the report, in "4. Settings".
- **Data Clock color:** red (default), blue, green, orange, purple, gray or teal. Affects the matrix, the bands block and the time-band chart.
- **Language:** Spanish, English or Portuguese, chosen from the "Language" menu, without restarting the program. The chosen language is saved and restored the next time the program is opened.

11. Export

- **PDF:** one file with every page of the report (Data Clock, by day, one per enabled analysis, and the General Summary if enabled). The number of pages varies depending on how many analyses were enabled.
- **Excel:** a "1. Data Clock" sheet with the full matrix, one sheet per enabled counting analysis (with the full ranking, no item limit), and a "General Summary" sheet if enabled.
- **PNG:** one image file per page of the report.

Files are suggested under the name "DataClock_YYYYMMDD_HHMMSS" (the date and time at the moment of export). After exporting, the program asks whether to open the generated file.

12. Save and load configuration

From the File menu or from the "1. Import" tab (buttons enabled once a file is imported), the whole current configuration can be saved to a .json file: the mapped columns, the filters (including each one's checked values) and the step-4 settings. This saves you from rebuilding everything when a new file of the same type (same columns) comes in.

When loading a saved configuration, the program validates every column against the currently imported file: if any no longer exists (e.g. the new file uses a different column name), it warns which ones with a message, instead of failing silently or applying something incorrect.

13. Frequently asked questions

Why doesn't the matrix total match the row count of my original file?

Because filters were applied, or because some rows were excluded for exceeding 24 hours in duration or for having a date/time format error. The text "Events analyzed: N · Excluded for exceeding 24 hs: M" shown below the Data Clock's title shows both numbers; $N + M$ should match the number of rows that passed the chosen filters.

Why does an analysis ask for a column I already picked?

Counting analyses use directly the column mapped in step 2 (Modality, DelitoCOP, Unit or Zone). If that message appears, it's because that specific column was left empty in step 2.

My DBF dates aren't importing correctly?

Try turning on "Manually choose the date and time format" in step 2, and pick the YYYYMMDD format for the date (that's DBF's native "Date" field format, which the program already converts to YYYY-MM-DD when reading it, but it's available as a manual option just in case).